

Q.1₁₁₃ The first 12 letters of the english alphabets are written down at random. The probability that there are 4 letters between A & B is :

(A) $\frac{7}{33}$

(B) $\frac{12}{33}$

(C) $\frac{14}{33}$

(D*) $\frac{7}{66}$

[Hint: $n(S) = 12!$; $n(A) = {}^{10}C_4 \cdot 4! \cdot 2! \cdot 7!$

$\therefore p = \frac{7}{66}$]

Q.2₂₂₃ Events A and C are independent. If the probabilities relating A, B and C are $P(A) = \frac{1}{5}$; $P(B) = \frac{1}{6}$; $P(A \cap C) = \frac{1}{20}$; $P(B \cup C) = \frac{3}{8}$ then

(A*) events B and C are independent

(B) events B and C are mutually exclusive

(C) events B and C are neither independent nor mutually exclusive

(D) events B and C are equiprobable

[Hint: $P(A \cap C) = P(A) \cdot P(C)$

$$\frac{1}{20} = \frac{1}{5} \cdot P(C) \Rightarrow P(C) = \frac{1}{4}$$

$$\text{now } P(B \cup C) = \frac{1}{6} + \frac{1}{4} - P(B \cap C), \text{ hence } P(B \cap C) = \frac{3}{8} - \frac{1}{3} = \frac{1}{24} = P(B) \cdot P(C) \Rightarrow (A)]$$

Q.3 Assume that the birth of a boy or girl to a couple to be equally likely, mutually exclusive, exhaustive and independent of the other children in the family. For a couple having 6 children, the probability that their "three oldest are boys" is

(A) $\frac{20}{64}$

(B) $\frac{1}{64}$

(C) $\frac{2}{64}$

(D*) $\frac{8}{64}$

[Hint: $E : B_1 B_2 B_3 \times \times \times$ where $\times$ means B or G

$$P(E) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8} = \frac{8}{64} \text{ Ans. }]$$

Q.4 A and B play a game. A is to throw a die first, and is to win if he throws 6, If he fails B is to throw, and to win if he throws 6 or 5. If he fails, A is to throw again and to win with 6 or 5 or 4, and so on, find the

chance of each player. [Ans. A : $\frac{169}{324}$; B : $\frac{155}{324}$]

Q.5₆₉ Box A contains 3 red and 2 blue marbles while box B contains 2 red and 8 blue marbles. A fair coin is tossed. If the coin turns up heads, a marble is drawn from A, if it turns up tails, a marble is drawn from bag B. The probability that a red marble is chosen, is

(A) $\frac{1}{5}$

(B*) $\frac{2}{5}$

(C) $\frac{3}{5}$

(D) $\frac{1}{2}$

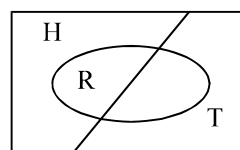
[Hint: A $\begin{cases} 3 \text{ R} \\ 2 \text{ B} \end{cases}$; B $\begin{cases} 2 \text{ R} \\ 8 \text{ B} \end{cases}$

R = event that a red marble is drawn

$$P(R) = P(R \cap H) + P(R \cap T)$$

$$= P(H) P(R/H) + P(T) \cdot P(R/T)$$

$$= \frac{1}{2} \left[\frac{3}{5} + \frac{2}{10} \right] = \frac{8}{10} \cdot \frac{1}{2} = \frac{2}{5}]$$



Q.6₇₄ A examination consists of 8 questions in each of which one of the 5 alternatives is the correct one. On the

assumption that a candidate who has done no preparatory work chooses for each question any one of the five alternatives with equal probability, the probability that he gets more than one correct answer is equal to :

- (A) $(0.8)^8$ (B) $3(0.8)^8$ (C) $1 - (0.8)^8$ (D*) $1 - 3(0.8)^8$

[Hint: $p = \frac{1}{5} = 0.2$; $q = 0.8$; $P(E) = 1 - P(0 \text{ or } 1)$]

Q.7₁₈₁ The germination of seeds is estimated by a probability of 0.6. The probability that out of 11 sown seeds exactly 5 or 6 will spring is :

- (A*) $\frac{{}^{11}C_5 \cdot 6^5}{5^{10}}$ (B) $\frac{{}^{11}C_6 (3^5 2^5)}{5^{11}}$ (C) ${}^{11}C_5 \left(\frac{5}{6}\right)^{11}$ (D) none of these

Q.8₁₅₂ The probability of obtaining more tails than heads in 6 tosses of a fair coins is :

- (A) $2/64$ (B*) $22/64$ (C) $21/64$ (D) none

[Hint: $P(4 \text{ or } 5 \text{ or } 6) = \frac{{}^6C_4 + {}^6C_5 + {}^6C_6}{64} = \frac{22}{64}$]

Q.9₁₉₂ An instrument consists of two units. Each unit must function for the instrument to operate. The reliability of the first unit is 0.9 & that of the second unit is 0.8. The instrument is tested & fails. The probability that "only the first unit failed & the second unit is sound" is :

- (A) $1/7$ (B*) $2/7$ (C) $3/7$ (D) $4/7$

[Hint:

A : the instrument has failed

B_1 : first unit fails and second is healthy

B_2 : first unit healthy and second unit fails

B_3 : both fails

B_4 : both healthy

$$P(B_1) = 0.1 \times 0.8 = 0.08$$

$$P(B_2) = 0.2 \times 0.9 = 0.18$$

$$P(B_3) = 0.1 \times 0.2 = 0.02$$

$$P(B_4) = 0.9 \times 0.8 = 0.72$$

$$P(A/B_1) = P(A/B_2) = P(A/B_3) = 1$$

$$P(A/B_4) = 0$$

Now compute $P(B_1/A)$]

Q.10₁₉₆ Lot A consists of 3G and 2D articles. Lot B consists of 4G and 1D article. A new lot C is formed by taking 3 articles from A and 2 from B. The probability that an article chosen at random from C is defective, is

- (A) $\frac{1}{3}$ (B) $\frac{2}{5}$ (C*) $\frac{8}{25}$ (D) none

[Hint: A = event that the item came from lot A ; $P(A) = \frac{3}{3+2} = \frac{3}{5}$

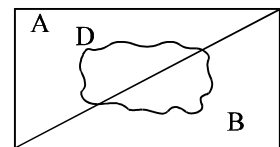
B = item came from B ; $P(B) = 2/5$

D = item from mixed lot 'C' is defective

$$P(D) = P(D \cap A) + P(D \cap B)$$

$$= P(A) \cdot P(D/A) + P(B) \cdot P(D/B)$$

$$= \frac{3}{5} \times \frac{2}{5} + \frac{2}{5} \times \frac{1}{5} = \frac{8}{25} \text{ Ans.]}$$



Q.11₂₈ A die is weighted so that the probability of different faces to turn up is as given :

Number	1	2	3	4	5	6
Probability	0.2	0.1	0.1	0.3	0.1	0.2

If $P(A/B) = p_1$ and $P(B/C) = p_2$ and $P(C/A) = p_3$ then the values of p_1, p_2, p_3 respectively are
Take the events A, B & C as $A = \{1, 2, 3\}$, $B = \{2, 3, 5\}$ and $C = \{2, 4, 6\}$

- (A) $\frac{2}{3}, \frac{1}{3}, \frac{1}{4}$ (B) $\frac{1}{3}, \frac{1}{3}, \frac{1}{6}$ (C) $\frac{1}{4}, \frac{1}{3}, \frac{1}{6}$ (D*) $\frac{2}{3}, \frac{1}{6}, \frac{1}{4}$

Q.12 If mn coins have been distributed into m purses, n into each find

- (1) the chance that two specified coins will be found in the same purse, and
(2) what the chance becomes when r purses have been examined and found not to contain either of the specified coins.

[Ans. (1) $\frac{n-1}{mn-1}$, (2) $\frac{n-1}{mn-rn-1}$]

Q.13 A box has four dice in it. Three of them are fair dice but the fourth one has the number five on all of its faces. A die is chosen at random from the box and is rolled three times and shows up the face five on all the three occasions. The chance that the die chosen was a rigged die, is

- (A) $\frac{216}{217}$ (B) $\frac{215}{219}$ (C*) $\frac{216}{219}$ (D) none

[Sol. $\begin{array}{l} 4 \leftarrow \begin{array}{l} 3 \text{ normal die} \\ 1 \text{ rigged die} \end{array} \end{array}$

A : die shows up the face 5

B_1 : it is a rigged die ; $P(B_1) = 1/4$

B_2 : it is a normal die ; $P(B_2) = 3/4$

$$P(A/B_1) = 1 ; P(A/B_2) = \frac{1}{216}$$

$$P(B_1/A) = \frac{\frac{1}{4} \cdot 1}{\frac{1}{4} \cdot 1 + \frac{3}{4} \cdot \frac{1}{216}} = \frac{216}{219} \text{ Ans.]}$$

Q.14 On a Saturday night 20% of all drivers in U.S.A. are under the influence of alcohol. The probability that a driver under the influence of alcohol will have an accident is 0.001. The probability that a sober driver will have an accident is 0.0001. If a car on a Saturday night smashed into a tree, the probability that the driver was under the influence of alcohol, is

- (A) $3/7$ (B) $4/7$ (C*) $5/7$ (D) $6/7$

[Hint: A : car met with an accident

B_1 : driver was alcoholic, $P(B_1) = 1/5$

B_2 : driver was sober, $P(B_2) = 4/5$

$P(A/B_1) = 0.001$; $P(A/B_2) = 0.0001$

$$P(B_1/A) = \frac{(.2)(.001)}{(.2)(.001) + (.8)(.0001)} = 5/7 \text{ Ans.]}$$

Direction for Q.15 to Q.17 (3 Questions)

A JEE aspirant estimates that she will be successful with an 80 percent chance if she studies 10 hours per day, with a 60 percent chance if she studies 7 hours per day and with a 40 percent chance if she studies 4 hours per day. She further believes that she will study 10 hours, 7 hours and 4 hours per day with probabilities 0.1, 0.2 and 0.7, respectively

- Q.15 The chance she will be successful, is
 (A) 0.28 (B) 0.38 (C*) 0.48 (D) 0.58

- Q.16 Given that she is successful, the chance she studied for 4 hours, is
 (A) $\frac{6}{12}$ (B*) $\frac{7}{12}$ (C) $\frac{8}{12}$ (D) $\frac{9}{12}$

- Q.17 Given that she does not achieve success, the chance she studied for 4 hour, is
 (A) $\frac{18}{26}$ (B) $\frac{19}{26}$ (C) $\frac{20}{26}$ (D*) $\frac{21}{26}$

[Sol. A : She get a success

[18-12-2005, 12 & 13]

T : She studies 10 hrs : $P(T) = 0.1$

S : She studies 7 hrs : $P(S) = 0.2$

F : She studies 4 hrs : $P(F) = 0.7$

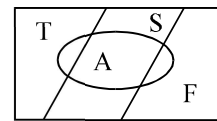
$P(A/T) = 0.8$; $P(A/S) = 0.6$; $P(A/F) = 0.4$

$P(A) = P(A \cap T) + P(A \cap S) + P(A \cap F)$

$= P(T) \cdot P(A/T) + P(S) \cdot P(A/S) + P(F) \cdot P(A/F)$

$= (0.1)(0.8) + (0.2)(0.6) + (0.7)(0.4)$

$= 0.08 + 0.12 + 0.28 = 0.48$ **Ans.(15)**



$$P(F/A) = \frac{P(F \cap A)}{P(A)} = \frac{(0.7)(0.4)}{0.48} = \frac{0.28}{0.48} = \frac{7}{12} \quad \text{Ans.(16)}$$

$$P(F/\bar{A}) = \frac{P(F \cap \bar{A})}{P(\bar{A})} = \frac{P(F) - P(F \cap A)}{0.52} = \frac{(0.7) - 0.28}{0.52} = \frac{0.42}{0.52} = \frac{21}{26} \quad \text{Ans.(17)]}$$

- Q.18 There are four balls in a bag, but it is not known of what colour they are ; one ball is drawn at random and found to be white. Find the chance that all the balls are white. Assume all number of white ball in the bag to be equally likely. [Ans. 2/5]

- Q.19 A letter is known to have come either from London or Clifton. On the postmark only the two consecutive letters ON are legible. What is the chance that it came from London? [Ans. 12/17]

- Q.20 A purse contains n coins of unknown value, a coin drawn at random is found to be a rupee, what is the chance that is it the only rupee in the purse? Assume all numbers of rupee coins in the purse is equally likely.

$$[\text{Ans. } \frac{2}{n(n+1)}]$$

- Q.21 One of a pack of 52 cards has been lost, from the remainder of the pack two cards are drawn and are found to be spades, find the chance that the missing card is a spade. [Ans. 11/50]

- Q.22 A, B are two inaccurate arithmeticians whose chance of solving a given question correctly are $(1/8)$ and $(1/12)$ respectively. They solve a problem and obtained the same result. If it is 1000 to 1 against their making the same mistake, find the chance that the result is correct. [Ans. 13/14]

[Hint: A : they obtained the same result

$$\left. \begin{aligned}
 B_1 : A \cap \bar{B} ; P(B_1) &= \frac{1}{18} \cdot \frac{11}{12} \\
 B_2 : \bar{A} \cap B ; P(B_2) &= \frac{7}{8} \cdot \frac{1}{12} \\
 B_3 : A \cap B ; P(B_3) &= \frac{1}{8} \cdot \frac{1}{12} \\
 B_4 : \bar{A} \cap \bar{B} ; P(B_4) &= \frac{7}{8} \cdot \frac{11}{12}
 \end{aligned} \right\}$$

$$\text{Now } P(A/B_1) = 0$$

$$P(A/B_2) = 0$$

$$P(A/B_3) = 1$$

$$P(A/B_4) = \frac{7}{8} \cdot \frac{11}{12} \cdot \frac{1}{1001}$$

$$\text{Now } P(B_3/A) = \frac{\frac{1}{8} \cdot \frac{1}{12}}{\frac{1}{8} \cdot \frac{1}{12} + \frac{7}{8} \cdot \frac{11}{12} \cdot \frac{1}{1001}} = \frac{13}{14}]$$

Q.23 We conduct an experiment where we roll a die 5 times. The order in which the number read out is important.

- (a) What is the total number of possible outcomes of this experiment?
- (b) What is the probability that exactly 3 times a "2" appears in the sequence (say event E)?
- (c) What is the probability that the face 2 appears at least twice (say event F)?
- (d) Which of the following are true : $E \subset F$, $F \subset E$?
- (e) Compute the probabilities : $P(E \cap F)$, $P(E/F)$, $P(F/E)$
- (f) Are the events E and F independent?

[Hint: (a) 6^5 ; (b) $P(E) = \frac{{}^5C_3 \cdot 5^2}{6^5} = \frac{10 \cdot 5^2}{6^5}$

(c) $P(F) = 1 - P(\text{face two appears exact once or no two}) = 1 - \frac{{}^5C_1 \cdot 5^4 + 5^5}{6^5}$; $P(F) = 1 - 2 \cdot \left(\frac{5}{6}\right)^5$

(d) $E \subset F$; (e) $P(E \cap F) = P(E)$; $P(E/F) = \frac{P(E \cap F)}{P(F)} = \frac{P(E)}{P(F)}$; $P(F/E) = \frac{P(E \cap F)}{P(E)} = 1$

(f) No]